

## Pure Bilinear Computational Results

The displayed formulation below is included only so that this file compiles independently. Copy only the block marked “BEGIN PAPER INSERT” into paper/Lorentz.tex.

$$\min\{c^\top x + d^\top y + y^\top R x : x \in \mathcal{E}_x, y \in \mathcal{E}_y\}. \quad (1)$$

We generated 120 random instances of (1) with  $\mathcal{E}_x = \{x \in \mathbb{R}^m : \|x\| \leq 1\}$  and  $\mathcal{E}_y = \{y \in \mathbb{R}^n : \|y\| \leq 1\}$ . For each candidate instance,  $R \in \mathbb{R}^{n \times m}$  was sampled with independent standard normal entries. Let  $u_1 \in \mathbb{R}^n$  and  $v_1 \in \mathbb{R}^m$  be the leading left and right singular vectors of  $R$ . We then set  $c = \alpha v_1 + \sigma g$  and  $d = \alpha u_1 + \sigma h$ , where  $g$  and  $h$  have independent standard normal entries,  $\alpha = 3.0$ , and  $\sigma = 0.1$ . Finally, the triple  $(c, d, R)$  was divided by  $\max\{\|c\|, \|d\|\}$ , so that  $\max\{\|c\|, \|d\|\} = 1$ . For each size we scanned seeds starting from zero and selected the first ten candidates for which the Shor relative gap  $(z_{\text{exact}} - z_{\text{Shor}})/|z_{\text{Shor}}|$  exceeded 0.01. We compared the Shor SDP relaxation, Full SEP, Lazy SEP, the Dual LOP/TRS bisection method, and Gurobi. Gurobi was run after the three conic methods with explicit bounds  $-1 \leq x_j \leq 1$ ,  $-1 \leq y_i \leq 1$  for each  $i$  and  $j$ , the two ball constraints, and time limit  $\max\{5, 3t_{\text{max}}\}$ , where  $t_{\text{max}}$  is the maximum time used by the three conic methods on that instance. Lazy SEP added at most 50 violated coordinate skew-skew equalities in each outer-approximation round, using violation tolerance  $10^{-7}$ . The Dual LOP/TRS method used bisection tolerance  $10^{-6}$ .

Table 1: Median wall-clock times for bilinear instances.

| $n \times m$   | cases | Shor | Full SEP | Lazy SEP | LOP/TRS | Gurobi    | Gurobi opt |
|----------------|-------|------|----------|----------|---------|-----------|------------|
| $2 \times 2$   | 10    | 1 ms | 1 ms     | 1 ms     | 18 ms   | 25 ms     | 10         |
| $4 \times 4$   | 10    | 1 ms | 2 ms     | 5 ms     | 23 ms   | 1.81 s    | 9          |
| $6 \times 6$   | 10    | 1 ms | 11 ms    | 21 ms    | 24 ms   | 5.00 s    | 0          |
| $8 \times 8$   | 10    | 2 ms | 49 ms    | 82 ms    | 28 ms   | 5.00 s    | 0          |
| $10 \times 10$ | 10    | 2 ms | 321 ms   | 318 ms   | 31 ms   | 13.29 s   | 0          |
| $15 \times 15$ | 10    | 3 ms | 12.00 s  | 3.10 s   | 40 ms   | 100.67 s  | 0          |
| $20 \times 20$ | 10    | 5 ms | 481.33 s | 51.20 s  | 60 ms   | 1444.06 s | 0          |
| $4 \times 8$   | 10    | 1 ms | 7 ms     | 16 ms    | 27 ms   | 5.00 s    | 0          |
| $8 \times 12$  | 10    | 2 ms | 248 ms   | 267 ms   | 33 ms   | 8.77 s    | 0          |
| $10 \times 20$ | 10    | 3 ms | 7.14 s   | 2.24 s   | 51 ms   | 77.54 s   | 0          |
| $15 \times 20$ | 10    | 4 ms | 59.18 s  | 15.22 s  | 54 ms   | 249.99 s  | 0          |
| $15 \times 25$ | 10    | 5 ms | 249.54 s | 24.95 s  | 65 ms   | 748.66 s  | 0          |

The median times required by different methods on these test instances are given in Table 1. In the table, the column  $n \times m$  gives the dimensions of  $y$  and  $x$ , respectively. Ten cases were run for each size. The Shor, Full SEP, Lazy SEP, LOP/TRS, and Gurobi columns give median wall-clock times for the five methods. The Gurobi opt column gives the number of instances solved to global optimality by Gurobi within its adaptive time limit.

In Table 2 we give additional numerical diagnostics for the same instances as in Table 1. By construction none of the Shor instances has a lower bound that agrees with the exact conic value within  $10^{-5}$ . The “Shor max gap” column gives the largest gap between the exact conic value and the Shor lower bound. The “Conic max diff” column is the largest objective difference among all the exact conic methods over all instances of that size. The Lazy SEP columns give the median/maximum outer-approximation rounds, total added coordinate skew-skew cuts and the largest final coordinate skew-skew equation violation. The LOP/TRS columns report the median/maximum TRS separation oracle solves for the Dual LOP/TRS method as well as the maximum final bisection bracket width.

Table 2: Additional diagnostics for bilinear instances.

| $n \times m$   | Shor                | Conic               | Lazy SEP |           |                      | LOP/TRS |                     |
|----------------|---------------------|---------------------|----------|-----------|----------------------|---------|---------------------|
|                | max gap             | max diff            | rounds   | cuts      | max skew             | oracles | max gap             |
| $2 \times 2$   | $2.9 \cdot 10^{-1}$ | $5.5 \cdot 10^{-8}$ | 2/2      | 1/1       | $2.3 \cdot 10^{-14}$ | 23/23   | $8.9 \cdot 10^{-7}$ |
| $4 \times 4$   | $2.4 \cdot 10^{-1}$ | $3.2 \cdot 10^{-8}$ | 2/2      | 36/36     | $5.8 \cdot 10^{-12}$ | 23/23   | $8.3 \cdot 10^{-7}$ |
| $6 \times 6$   | $1.9 \cdot 10^{-1}$ | $1.2 \cdot 10^{-7}$ | 3/4      | 100/150   | $1.3 \cdot 10^{-8}$  | 23/23   | $8.1 \cdot 10^{-7}$ |
| $8 \times 8$   | $1.4 \cdot 10^{-1}$ | $3.2 \cdot 10^{-8}$ | 5/6      | 200/250   | $2.2 \cdot 10^{-9}$  | 23/23   | $7.5 \cdot 10^{-7}$ |
| $10 \times 10$ | $1.2 \cdot 10^{-1}$ | $4.1 \cdot 10^{-8}$ | 9/12     | 400/550   | $2.1 \cdot 10^{-9}$  | 23/23   | $7.3 \cdot 10^{-7}$ |
| $15 \times 15$ | $6.8 \cdot 10^{-2}$ | $1.2 \cdot 10^{-8}$ | 22/37    | 1025/1800 | $3.0 \cdot 10^{-9}$  | 23/23   | $6.7 \cdot 10^{-7}$ |
| $20 \times 20$ | $6.7 \cdot 10^{-2}$ | $5.5 \cdot 10^{-9}$ | 63/76    | 3100/3750 | $9.9 \cdot 10^{-9}$  | 23/23   | $6.6 \cdot 10^{-7}$ |
| $4 \times 8$   | $1.8 \cdot 10^{-1}$ | $3.2 \cdot 10^{-7}$ | 3/4      | 100/150   | $1.8 \cdot 10^{-9}$  | 23/23   | $7.9 \cdot 10^{-7}$ |
| $8 \times 12$  | $1.1 \cdot 10^{-1}$ | $1.2 \cdot 10^{-7}$ | 9/11     | 400/500   | $2.2 \cdot 10^{-9}$  | 23/23   | $7.1 \cdot 10^{-7}$ |
| $10 \times 20$ | $8.7 \cdot 10^{-2}$ | $8.5 \cdot 10^{-9}$ | 20/30    | 950/1450  | $2.2 \cdot 10^{-9}$  | 23/23   | $6.8 \cdot 10^{-7}$ |
| $15 \times 20$ | $6.0 \cdot 10^{-2}$ | $2.3 \cdot 10^{-8}$ | 41/54    | 2000/2650 | $1.9 \cdot 10^{-9}$  | 23/23   | $6.5 \cdot 10^{-7}$ |
| $15 \times 25$ | $5.5 \cdot 10^{-2}$ | $3.1 \cdot 10^{-7}$ | 46/82    | 2250/4050 | $2.3 \cdot 10^{-9}$  | 23/23   | $6.5 \cdot 10^{-7}$ |

To summarize these results, Lazy SEP is slower than Full SEP on smaller instances due to the overhead of checking for violated skew-skew constraints and repeated conic solves, but the time for Full SEP blows up faster as problem size increases. The Dual LOP/TRS bisection algorithm is very fast and robust and scales much better than Full SEP or Lazy SEP. These problems are difficult for Gurobi, even with the explicit variable bounds added, and cannot be solved in time competitive to the conic methods.